808
EQ+

EXPANDED TS808-STYLE OVERDRIVE

USER MANUAL

A beginner-friendly guide to connections, controls, sound modes, and everyday operation.

DRAFT EDITION | 9 V DC CENTER-NEGATIVE | TRUE BYPASS

01

Quick Start

Connect the pedal safely, choose a familiar starting sound, and begin at a controlled volume.

1. **Turn the volume down**
   Lower the amplifier or interface input before making connections.
2. **Connect your guitar**
   Use the right-side **IN** jack when the pedal is viewed from above.
3. **Connect the output**
   Run the left-side **OUT** jack to the next pedal or amplifier.
4. **Connect power**
   Use regulated **9 V DC, center-negative** power or a correctly wired 9 V battery adapter.
5. **Choose a starting point**
   Set Drive and Tone near noon, Level low, and select Tight + Sym with both Cut settings.
6. **Engage and adjust**
   Press the footswitch. The blue LED turns on; raise Level slowly while playing.

SIGNAL FLOW

GUITAR → IN | 808 EQ+ | OUT → NEXT PEDAL → AMPLIFIER

Power: regulated 9 V DC supply or correctly wired 9 V battery

POWER CHECK

Use a regulated **9 V DC, center-negative** supply, or a 9 V battery connected through a center-negative 2.1 mm plug. Battery power is portable, but verify the connector polarity before use. Do not apply more than 9 V.

Your first useful check

With the effect engaged, move Level until the amplifier is about as loud as it is when the pedal is **off** - meaning the effect and LED are deactivated. This matching point is called **unity gain**. From there, Drive, Tone, and switch changes are easier to compare because a volume difference is not dominating what you hear.

START GENTLY
Open mode can produce substantially more output than Tight mode. Lower Level before changing clipping modes at high listening volume.

02

Pedal at a Glance

1. **9 V DC**
   Regulated center-negative power
2. **Drive**
   Overdrive intensity
3. **Tone**
   Dark to bright
4. **Level**
   Final output volume
5. **Tight / Open**
   Compressed or dynamic clipping
6. **Sym / Asym**
   Tight-mode clipping texture
7. **Cut / Treb**
   Traditional or extended clarity
8. **Cut / Bass**
   Traditional or extended fullness
9. **Status LED**
   On while effect is active
10. **Footswitch**
    Effect / true bypass
11. **Input**
    Right-side 1/4-inch jack
12. **Output**
    Left-side 1/4-inch jack

Arrow reminder: Sym/Asym only changes the sound when Tight is selected. In Open mode, that switch is ignored.

03

Controls in Plain Language

Start with what each control feels like under your fingers; the circuit details can wait.

DRIVE

Controls distortion. Turning clockwise adds grit, saturation, compression, and sustain. Lower settings can add only a slight edge or help push an amplifier.

TONE

Counterclockwise is darker and smoother; the extreme can sound muddy and muted. Clockwise adds presence and brightness; the extreme can sound harsh or fizzy.

LEVEL

Sets final output volume. Tune it to unity gain, or raise it above unity to drive the next device harder.

2 clipping switches, 2 EQ switches

TIGHT	Focused, smooth, and more compressed	**OPEN**	Louder, more dynamic, and less compressed
SYM	Symmetric clipping: classic and even	**ASYM**	Asymmetric clipping: thicker and grittier
CUT	Traditional smoother treble response	**TREB**	More clarity, presence, and sparkle
CUT	Traditional tight low end	**BASS**	More fullness and weight

Easy classic reset: flip all four switches upward. This selects Tight + Sym with both EQ switches at Cut, restoring the familiar TS808-style voicing and clipping behavior.

04

Sound Modes at a Glance

Think of the switches as three layers: clipping feel, clipping texture, and frequency range.

TIGHT + SYM

Classic and focused. Smooth, compressed, sustained, and familiar. A natural choice for tightening and pushing a dirty amplifier.

TIGHT + ASYM

Textured and thicker. Slightly louder and less even, with extra grit and a more vintage or amp-like edge.

OPEN

Dynamic and spacious. Louder, less compressed, and more responsive to picking strength. Sym/Asym is ignored.

Shape the range

TREB CUT

Smoother and more focused; closer to the traditional Tube Screamer upper-frequency response.

TREB

Clearer, brighter, more present, and more open. Lower Tone if the result becomes sharp.

BASS CUT

Tighter and leaner. Helpful with high Drive, bass-heavy amplifiers, or dense mixes.

BASS

Fuller and thicker, adding weight beneath the distorted sound.

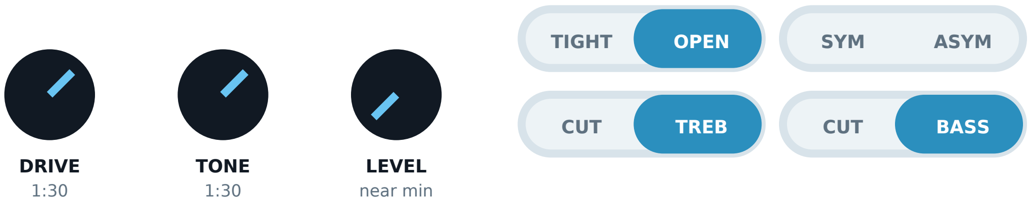
Combine freely. Bass and Treb operate independently. Use either extension alone, both together, or neither. The best combination depends on the guitar, amplifier, speaker, and how hard you are driving the pedal.

05

Three Places to Begin

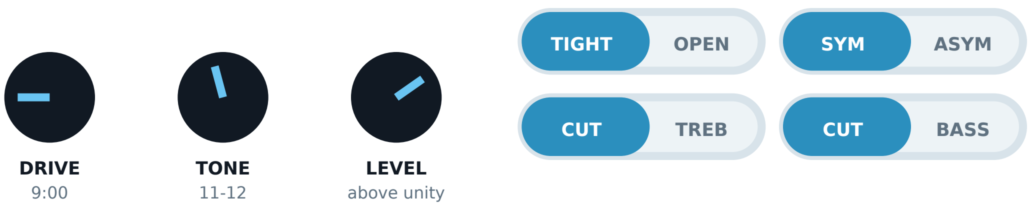
Approximate clock positions are only starting points. Match Level to your rig before judging the tone.

01 Dynamic and Responsive



The designer's favorite: play softly for a cleaner response and dig in for grit. Adjust Level carefully until the effect-on and pedal-off volumes are similar.

02 Traditional TS808-Style Boost



Tightens the low end, adds smooth clipping, and pushes an amplifier that is already beginning to distort. Raise Level above unity to drive the amp harder.

03 EXPLORE THE EXTREMES

Try Treb with Tone fully clockwise for a sharp, immediate sound, or select Bass and turn Drive high for a larger, heavier character. Compare pickup selections and change how hard you play. The setting that makes you keep playing is the right one.

06

Why the Controls Change the Sound

The circuit is complex, but its musical behavior can be understood without equations.

DRIVE | HOW HARD IT IS PUSHED

Drive enlarges the signal entering the clipping section. As the signal grows, its peaks are rounded and limited more strongly, creating overdrive, compression, and sustain.

TONE | UPPER-MID AND TREBLE SHAPE

Tone primarily controls upper-mid and high frequencies while leaving low and lower-mid content largely unchanged. The Treb switch sets the available high-frequency range before the signal reaches Tone.

BASS | WEIGHT INTO THE OVERDRIVE

Bass allows more low-frequency content to receive distortion gain. The result is fuller and larger. Bass Cut keeps low frequencies tighter and can preserve definition with high Drive or a bass-heavy rig.

TREB | DETAIL BEFORE THE TONE CONTROL

The Treb switch is implemented before the Tone control. Treb allows substantially more upper-frequency detail into that stage, increasing clarity and openness; Treb Cut restores the smoother traditional response.

The controls interact

A bright guitar through a bright amplifier may need less Tone or Treb. High Drive can make extra bass feel looser, while Open mode may require a lower Level setting. Treat the controls as a system rather than isolated adjustments.

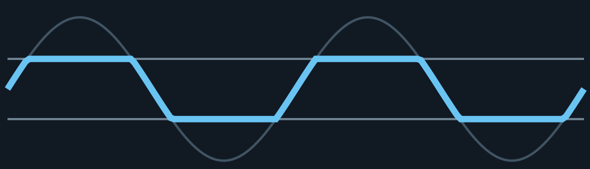
06

Clipping Feel and Texture

Clipping changes more than the amount of distortion: it also changes compression, output, and touch response.

TIGHT

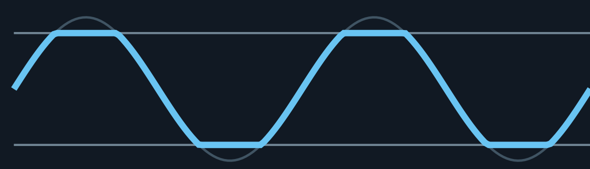
Silicon diodes begin shaping the signal at a lower level.



EARLIER CLIPPING | MORE COMPRESSION | MORE SUSTAIN

OPEN

LEDs require a larger signal before strongly shaping it.



MORE HEADROOM | MORE OUTPUT | MORE DYNAMICS

SYM | SYMMETRIC CLIPPING

In Tight mode, symmetric clipping shapes the positive and negative halves similarly. This creates the traditional smooth, balanced TS808-style character and tends to emphasize odd-order harmonics.

ASYM | ASYMMETRIC CLIPPING

In Tight mode, asymmetric clipping treats the two halves differently. It adds a little output and a thicker texture, with a combination of odd- and even-order harmonics.

Why the arrow is printed on the pedal: the Sym/Asym switch belongs to Tight mode. Open selects its own LED clipping arrangement, so moving Sym/Asym while Open is selected produces no change.

07

Power, Bypass, and Physical Details

The essentials for reliable everyday use and pedalboard integration.

POWER

9 V DC | center-negative | 2.1 mm

Average current draw is approximately 7-10 mA. A 9 V battery with a verified center-negative connection also works and is a portable option. Do not use higher voltage.

AUDIO AND BYPASS

Right-side 1/4-inch jack: **IN**
Left-side 1/4-inch jack: **OUT**

True bypass completely disconnects the guitar from the effect circuit while off, preserving the guitar's raw tone as much as possible.

Signal and enclosure

ACTIVE

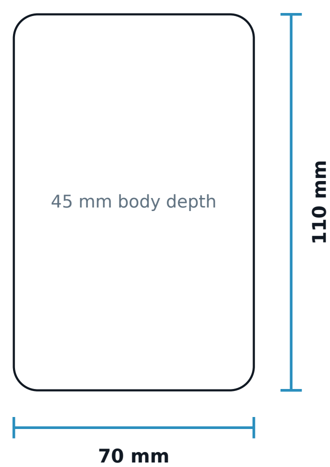
IN -> input buffer -> drive / clipping

-> tone / level -> output buffer -> OUT

BYPASS

IN ------------------------------------> OUT

LED on = active effect | LED off = true bypass



Approximate body size

EFFECT	TS808-style overdrive with expanded EQ and clipping
CONTROLS	Drive, Tone, Level; four two-position toggles
POWER	Regulated 9 V DC center-negative
BATTERY	External 9 V battery supported with verified center-negative 2.1 mm connection
CURRENT	Approximately 7-10 mA average
ENCLOSURE	3D-printed plastic; approximately 110 x 70 x 45 mm body

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Troubleshooting and Care

Start with connections, power, and control positions before assuming an internal fault.

NO SOUND

Confirm the guitar is connected to right-side IN, left-side OUT feeds the amplifier, and both cables work.

TOO BRIGHT

Lower Tone or select Treb Cut.

BYPASS WORKS; EFFECT DOES NOT

Check regulated 9 V center-negative power, confirm Level is not at minimum, and look for the active LED.

TOO MUDDY

Select Bass Cut, reduce Drive, or turn up Tone.

TOO LOUD

Lower Level. Open mode can produce much more output than Tight mode.

SYM/ASYM HAS NO EFFECT

Select Tight. Sym/Asym is intentionally ignored in Open mode.

STEADY NOISE

Try a known low-noise or isolated supply. The active circuit can amplify supply noise along with the guitar signal.

BRIEF SWITCH POP

A small pop can occur when a tone-switch capacitor changes. Switch at a sensible listening level.

PRINTED ENCLOSURE CARE

The printed enclosure does not provide the impact resistance, heat resistance, or inherent electromagnetic shielding of a grounded cast-metal enclosure. Avoid excessive heat, impact, crushing force, and unnecessary stress on the controls and connectors.

If a problem persists, disconnect the pedal from valuable equipment before inspecting or modifying it.

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KEEP PLAYING.
KEEP EXPLORING.

The best setting is the one that responds to your hands, your guitar, and your amplifier. Use the starting points, then make the pedal your own.

PROJECT DOCUMENTATION

Build guide | schematic | bill of materials | PCB and perfboard resources | CAD drawings | print files | technical documentation

github.com/mbovero/808_EQ-Plus_Guitar_Pedal

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